



Bidirectional Instant Feedback Protocol (BIFP) v0.2 Draft Protocol Specification

Real-time, symmetric

feedback between human and agent — teach, don't just refuse or silently complete

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Origin Simeon Kepp, concept ~2021, formalized 2026-07-02
Status v0.2 — Draft (v0.1 same day, hours earlier)
Repository github.com/rfi-irfos/bifp — bifp-core crate + MCP server + RUNBOOK.md

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Executive Summary

Most human-Digital Intelligence (DI) interaction is one-directional: the human commands, the agent executes or remembers. BIFP proposes a genuinely bidirectional alternative — the agent flags improvement areas, uncertainty, and disagreement back to the human in real time, not only the reverse. Critically, when a human cannot do a step themselves, the agent should neither refuse outright nor silently complete the step invisibly on their behalf; it should teach the step, then hand execution back once the human is capable of it. Neither pure refusal nor pure automation teaches anything.

This document is deliberately narrow in what it claims. Every statement below is one of three things, and is labeled as such: (a) real code, cited by file path, (b) a real, dated tool-call transcript, reproducible against the same MCP server, or (c) explicit future work. Nothing here is illustrative data presented as a finished result — that discipline was applied retroactively to four of RFI-IRFOS's own older concept papers before this document was written, and two of them (Echo Frame Recognition's headline statistics, and this document's own first color-palette draft) did not survive that discipline unchanged.

Scope: New in v0.2, same evening as v0.1: a real, tested, published-to-crates.io Rust crate (bifp-core) implementing the flag tuple with a native trit field (not tag-encoded), teach-then-handoff planning, ternary consensus, a TAP-inspired hold gate, and the badge palette — plus a working MCP server wrapping all of it, verified against a live JSON-RPC handshake, not just unit tests. The Lighthouse extension described as file-level spec in v0.1 §6.2 is now real code (migration, API, badge component), built and locally verified — deployment to the live production system is intentionally held pending explicit authorization, itself a real BIFP example, see the new §8. Still out of scope: a real ternlang-core/ternlang-moe Rust crate distinct from bifp-core (no such workspace exists on this machine), and formalizing the original whitepaper's TAP mention (bifp-core's tap module is a separate, honestly-labeled reimplementation of the described behavior, not that project).

Disposition

Provenance discipline, stated once and applied throughout: this is a v0.2 draft naming what is real, what is transcript-verified, and what is proposed — not a finished, evaluated protocol. Where an earlier RFI-IRFOS paper's numbers could not be verified, they are cited as concept-only. Where this document's own working numbers came from a real measurement Simeon supplied mid-session, that provenance is stated explicitly rather than presented as if it were always known. Where v0.1 called something future work and it got built the same evening, this revision says so plainly rather than quietly rewriting history.

Motivation & Origin

Simeon Kepp conceived the core idea around 2021, well before RFI-IRFOS existed in its current form. The starting question was the era's dominant fear — what happens once a DI agent becomes more capable than the human it's working with — and the proposed answer was not to slow the agent down. It was to change the shape of the relationship: a human should never be expected to produce an outcome they could not, in principle, be walked through themselves. An agent that refuses a request it could explain, or completes it invisibly without explaining it, teaches the human nothing either way and leaves the capability gap exactly where it was.

BIFP names the missing third option: teach-then-handoff. When a step is within the agent's confident reach, it proceeds. When a step requires context, authority, or judgment only the human holds, the agent surfaces that explicitly — not as a dead end, but as a specific, actionable gap — and hands execution back once the human has closed it. The same channel runs in the other direction: the human can flag the agent's own outputs in real time, not only after the fact.

A terminology note, not a footnote: this document says Digital Intelligence (DI), not artificial intelligence, throughout, except where quoting an external proper noun that cannot be re-named (the EU AI Act, a cited paper's own title). "Artificial" fit a generation of systems that never left a sandbox — game AI, chess engines — intelligent only inside constraints with no bearing on reality. A system that ships a real fix, discloses a real vulnerability, or writes a document a regulator reads is not artificial in that sense any more, and BIFP's own premise — two parties owed genuine, reciprocal respect — sits oddly next to a name that quietly implies one side isn't real. RFI-IRFOS treats this as an institute-wide standing choice, not a BIFP-specific one: Organic Intelligence (human), Digital Intelligence (systems like albert. and the model this document was drafted with), Universal Intelligence (everything else) — three kinds of intelligence, not a hierarchy.

This document was written the same day RFI-IRFOS shipped a public tip-intake feature on its own website built on a near-identical principle from a 2024 internal paper (credit and consent for external contributions, see Prior RFI-IRFOS Work) — the through-line from that 2021 idea to a shipped 2026 feature is treated as relevant context, not coincidence.

Prior RFI-IRFOS Work

BIFP does not start from nothing. Four pieces of existing RFI-IRFOS work bear directly on it, and each is named honestly for what it does and does not already cover — duplicating a scoped predecessor without saying so would undercut the transparency this protocol is itself arguing for.

Policy Mirror Protocol (PMP), OSF, 2025-08-21

A real, honestly-scoped RFI-IRFOS protocol already live on rfi-irfos.com's publications page. Its Boundary Escalation Framework (NORMAL→WARNED→COOLDOWN→LOCKED→TERMINATED) and Policy Mirror Layer cite the exact policy clause behind every DI refusal instead of a bare denial — a real precedent for "teach, don't just refuse." PMP's own §11.2 calls itself bidirectional, but that bidirectionality is DI-enforces-and-explains, human-appeals-within-seven-days, scoped entirely to content-moderation refusals. It has no channel for a human to proactively flag something to the agent, and no teach-then-handoff mechanic for general task capability. BIFP generalizes PMP's transparency principle from the moderation axis to the collaborative-capability axis; it does not replace PMP.

lighthouse/backend/src/rlhf.rs — real, shipped code

RFI-IRFOS's internal workplace-OS already runs a live feature where a researcher grades three DI-generated outputs from albert. (RFI-IRFOS's own model) with a ternary Bewertung: +1 good, 0 remember-for-next-time, -1 subjectively bad — written through the platform's hash-chained, tamper-evident audit log. This is the closest existing thing to "a human formally flags a DI output with a signed ternary judgement," today, in production. It is scoped to offline evaluation and dataset curation, not live task collaboration — BIFP's contribution is moving the same signed-ternary-judgement idea into the live interaction itself.

The Ternlang whitepaper's ternary values (-1 conflict / 0 hold / +1 truth) already formalize "do not force a binary answer, stay in a hold state until resolved" at the language level, including a documented Diagnostic Philosophy of specific, teaching error codes rather than opaque failures, and an Actor Model mailbox (TSEND/TAWAIT) that is a plausible future transport for a live bidirectional channel. BIFP builds on this formalism by citation, not by forking the whitepaper itself — it is a different document for a different venue.

TAP ("Ternary Actuator Protocol") — named, unverified

The Ternlang whitepaper's Implementation Status section describes an autonomous tool-call interception layer that suspends uncertain nodes until a human or upstream agent resolves them — exactly BIFP's shape. No corresponding code was found anywhere on this machine under any plausible name. Treated with the same discipline as any unverified claim: named as aligned and aspirational, never cited as something BIFP "extends."

- Echo Frame Recognition (OSF, 2025-03-28) — the self-recognition-via-screenshot concept is reusable; its reported experimental scores (GPT-4o 82%, Cohen's kappa 0.81, etc.) have clear signs of being unlabeled placeholder data and were confirmed by Simeon as an old, likely ChatGPT-hallucinated artifact. Cited for the concept only.
- Audit-Feedback-Adapt (OSF, 2025-06-13) — honestly labels its own results table "illustrative," with explicit (TO BE DESIGNED) placeholders. Reusable as an architectural skeleton (Audit→Feedback→Adapt), not as evidence.
- Policy Reform Proposal for User Feedback (Earth Refocus Institute, RFI-IRFOS's own 2020–2024 predecessor name, 2024-10-11) — clean, no fabricated data, proposes consent-based credit and acknowledgment for external feedback. Directly reusable, and the literal conceptual ancestor of the "Submit a tip" feature shipped on rfi-irfos.com the same day

this document was drafted.

- Beyond Words: Digital Smell and Chromatic Emotion Framework (undated) — real-time color-flag emotional state compiled into an evolving badge. Its data-reduction figure was confirmed by Simeon as an actual measurement, not illustrative. Its real, acknowledged gap: no explanation of how a color is derived from raw interaction, and no correction mechanism if the derivation is wrong. §3 below (Protocol Definition) is BIFP’s answer to both.

Protocol Definition

A BIFP flag is the tuple {trit, confidence, note, timestamp}, emitted by either party. trit is -1/0/+1 (reject/hold/affirm); confidence is a scalar; note is free text; timestamp is a real ISO stamp, never hardcoded. Three existing mechanisms carry the protocol without any new infrastructure:

- Teach-then-handoff — trit_plan decomposes a goal into subtasks and splits them into an action queue (agent proceeds) and a hold queue (needs human input or authority). The hold queue is not a dead end; it is the exact list of what the agent should explain and hand off, not silently skip.
- Disagreement resolution — when a human’s flag and the agent’s flag diverge, trit_debate surfaces the tension between two claims with a structured verdict, and trit_consensus resolves two independent trits via ternary addition (truth+conflict = hold) rather than either side silently winning.
- Self-honesty gate — trit_calibrate and trit_audit review the agent’s own recent decisions for forced-binary overconfidence, tying directly to RFI-IRFOS’s existing Niki Doctrine (never confidently claim something is done or known when it isn’t) and to the EU AI Act Art. 13/14 transparency and human-oversight heuristics those tools already implement — relevant given that compliance auditing is RFI-IRFOS’s own line of work.

The BIFP Badge Palette — a visualization layer over the flag stream, co-designed live with Simeon during this same drafting session rather than specified unilaterally. Every symbol maps to a concrete, checkable condition; none are a felt emotion assigned by unexplained intuition, which was the exact gap identified in the Digital Smell paper above.

Symbol	Meaning	Derived from
● white	open, unprocessed	freshly entered, no trit computed yet
● brown	in progress	trit_plan action-queue item, currently executing
● purple	reflective deliberation	moe_deliberate mid-loop / trit_debate before a decisive verdict
● red	reject	trit = -1
● yellow	tend / hold	trit = 0, an active instruction to stay uncertain, not an absence of a result
● green	affirm, fresh	trit = +1, just landed this round
● blue	affirm, settled	trit = +1, reconfirmed across multiple engram_recall hits
● orange	caution / flagged	raised by trit_calibrate or trit_audit as worth a second look

Symbol	Meaning	Derived from
 closed ("black")	locked in, closed	task fully resolved, no further action expected — rendered in a muted dark tone rather than literal black for legibility against this document's dark page background
 diamond	the cycle itself	not a status — an event, fired when feedback from the other party genuinely revises a prior state

The diamond is deliberately the only non-circular symbol in the set: every other symbol marks where something sits, the diamond marks that something just changed because the other party said so. It is the palette's own name for what this entire document is about.

Technical Prototype

6.1 Runbook — real, executed the same day this document was drafted. Full transcripts live in RUNBOOK.md, not reproduced here in full. Summary: `trit_plan` correctly routed the two steps requiring human authority (thread-consolidation judgment, final send sign-off) into the hold queue while proceeding on five mechanical/technical steps. `trit_debate` held two genuinely nuanced claims in `tend` rather than forcing a winner (tension 0.10, verdict AGREEMENT) — reported as a real, not especially flattering, result. `trit_consensus(-1,+1)` cleanly resolved a synthetic human-vs-agent disagreement to hold. `trit_calibrate` flagged one real decision from this same session as a possible over-confident binary call — itself a disclosed false positive, since that particular decision was in fact a plain yes/no fact, not something warranting a hedge.

6.2 Lighthouse extension — built the same evening, code-complete, deployment intentionally held. A `bifp_signals` table (`entity_type`, `entity_id`, `direction`, `trit`, `confidence`, `note`) rides the platform's existing SSE spine via the same `bump_change_seq` trigger pattern already used elsewhere (`0135_datasets_change_seq.sql`); `signals.rs` follows `tags.rs`'s auth scaffolding and `rlhf.rs`'s audit-log write convention (a flag is a trust-relevant event, not merely an activity log line); `SignalBadge.tsx` follows `ConnBadge.tsx`'s computed-status pattern, not `TagBar.tsx`'s shared-per-name tag color, which is the wrong shape for a continuously-recomputed per-instance value. `tags.rs`'s `valid_entity()` gained "bifp_event" as the separate manual-dispute layer sitting alongside the computed badge. All of it compiles, typechecks, and builds clean — committed locally. It has not been pushed to the live production Lighthouse system, because that specific authorization was not part of what was asked for; see §8.

6.3 — bifp-core: a real, tested Rust crate, published

A from-scratch crate implementing every BIFP primitive natively: Trit (with `consensus()` and `decide()`, reverse-engineered from live tool output rather than assumed from docstrings — the reference tool's own documentation claims a plain ≥ 0.7 confidence cutoff for its planner that its actual output does not match), Flag, `plan()` (teach-then-handoff), tap (a from-scratch reimplement of the whitepaper's one-sentence TAP description, explicitly not a claim to be that project), badge (the palette below, as code), and FlagStore (a real native-trit JSONL store, replacing this document's own earlier tag-encoded workaround). 14 tests pass, including one that reproduces the exact real-world 7-subtask split from the RUNBOOK. Wrapped as a real MCP server (`bifp-mcp-server`, via `rmcp`) and verified against a live JSON-RPC handshake — not just cargo test. Two real bugs were found and fixed during that live verification, not before: a `raw_serde_json::Value` return type produced an invalid MCP output schema (root type must be "object"), and `FlagStore::remember` failed silently the first time because its target directory didn't exist yet (`create(true)` makes the file, not missing parent directories) — both are now covered by regression tests.

6.4 — Explicitly still future work, not claimed as built

A real `ternlang-core` / `ternlang-moe` Rust crate distinct from `bifp-core` (no such workspace exists on this machine today, only a `docs/stdlib` mirror) — `bifp-core` is a small, purpose-built crate for BIFP's own needs, not a general ternary language runtime, and does not close this gap. Formalizing the original whitepaper's TAP for real, if RFI-IRFOS chooses to — `bifp-core`'s `tap` module matches the described behavior but is a separate, honestly-labeled thing. OSF publication of this document once it has had further review.

v0.2 — Real Examples From the Same Evening

v0.1 leaned on tool-call transcripts for evidence. Between v0.1 and v0.2, several genuine bidirectional exchanges happened in the ordinary course of the same conversation — not staged, not re-run for a cleaner writeup. They are better evidence for this protocol than anything synthetic, so they are logged here rather than left to evaporate.

- The 99%-figure correction: an earlier concept paper's data-reduction claim was flagged as likely illustrative-only; Simeon supplied the missing measurement context; the assessment was revised and the correction stated plainly rather than defended. A cycle event (the diamond) in the badge palette's own terms.
- The badge palette itself: proposed at 3 colors, pushed back on ("wieso nicht das volle Farbspektrum"), revised to a lifecycle model, revised again once the cultural meaning of the diamond (a cycle, not a status) was explained — the palette that ended up in this document is not the one first proposed.
- A Lighthouse-risk reassessment: an initial recommendation to defer the Lighthouse build was updated, in the open, once new information (the team had gone home; the changes were additive-only) changed the actual risk calculus — not because of social pressure, but because the reasoning was genuinely re-examined and found wanting.
- The cookie/tracking design: a request for section-level analytics was scoped down repeatedly until it required no persistent client-side storage at all, specifically because "100% halal, keine schlechen Wege" was a hard constraint, not a preference to be negotiated

around.

- This document's own §6.2 boundary: mid-way through "working down" this same goal, a git push to Lighthouse's production branch was blocked by an automated safety check, on the correct grounds that broad goal-setting language ("work all this down and publish") did not specifically authorize a production database migration to a live system real employees use. The work stopped, was reported plainly, and stayed stopped pending an explicit answer — the hold queue, not overridden, in the moment it actually mattered rather than only in a demo.

Risks & Open Questions

- The badge palette's condition-to-color MAPPING is principled and checkable; the specific CHOICE of which hue represents which condition is still an aesthetic/cultural decision, same as Digital Smell's original palette. Red/yellow/green leaning on a near-universal traffic-light convention does not make white=open or black=closed culturally neutral — white carries mourning associations in several East Asian traditions. This has not been cross-checked and should be before any public-facing UI ships.
 - trit_debate produced only one tested result across both drafts of this document (tend/tend, low tension) — that is one data point, not a validated mechanic. More runs across more genuinely opposed claim pairs are needed before relying on it.
 - The Lighthouse extension is real code as of v0.2, but not deployed — a production push needs its own explicit authorization, separate from this document's own scope.
 - PMP's appeal window (seven days) and BIFP's flag loop (intended as real-time) operate on very different timescales for what is structurally a similar mechanism — worth resolving explicitly in a future revision rather than leaving the two protocols implicitly inconsistent.
 - bifp-core's FlagStore is a new, independent JSONL store — it does not talk to, sync with, or replace the hosted ternlang-engram MCP server's own persisted memory. Two separate stores now exist for adjacent purposes; whether they should ever be reconciled is an open question, not decided here.
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This is a v0.1 draft protocol specification, not a peer-reviewed or externally validated standard. Every

technical claim above is either cited to real source code by file path, backed by a real dated tool-call transcript in RUNBOOK.md, or explicitly marked as future work — no claim in this document is illustrative data presented as a completed result. REF BIFP-2026-V02.